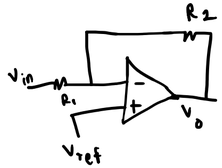
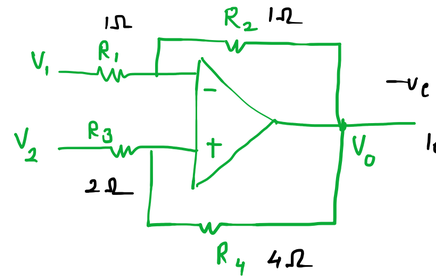
2. Op amp and Phasor

12/03/25 9:08 PM

Opamp Q ⇒ Use superposit^n

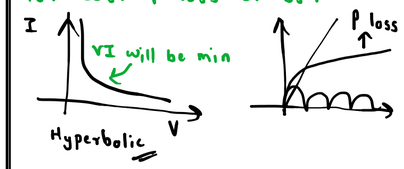


$$V_o = \underbrace{V_{in} \times \frac{-R_2}{R_1}}_{\substack{inv \\ V_{ref}=0}} + \underbrace{\left(1+\frac{R_2}{R_1}\right)V_{ref}}_{\substack{non\ inv \\ V_{in}=0}}$$



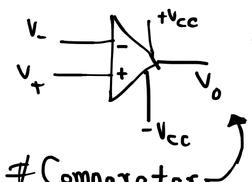
−ve feedback cuz less impea.

What kinda curve would we want for least P loss of BJT



Open loop gain

[virtual short is only in case of −ve feedback]

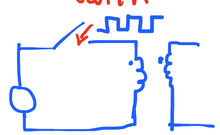


$V_o = A_o(V_+ - A_-)$

Comparator

Accumulator is 400V. But some other comp. require less than 400, 12V ex LED. HOW?

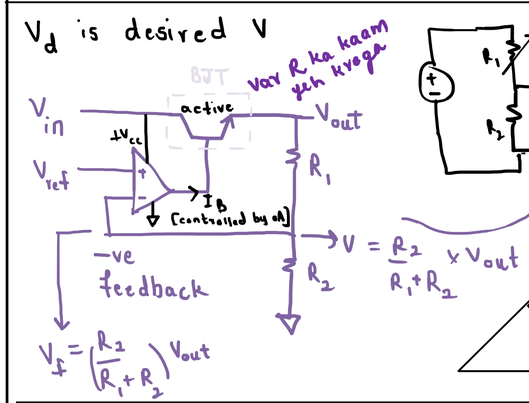
transformer? Nope cuz that needs AC V ↘ A fix is to use PWM at swith



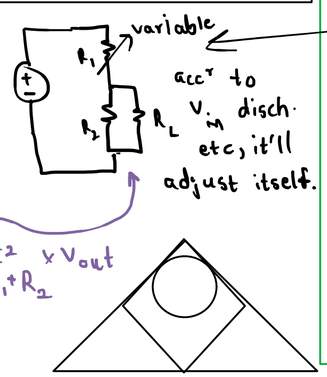
Opamp? +Vcc & −Vcc are also reqd

V divider? R1 & R2 but P loss & V discharges so V divided will also change.

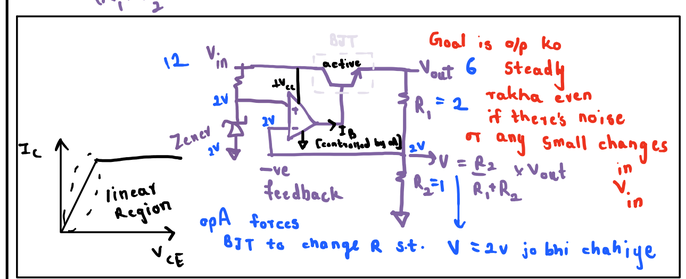
V_d is desired V



var R ka kaam yeh kregā



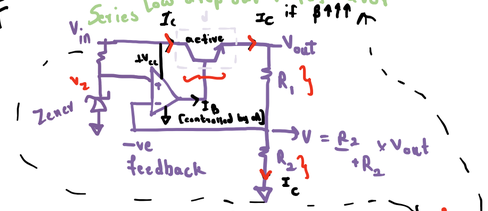
acc^r to V_{in} disch. etc, it'll adjust itself.



Goal is o/p ko steady rakha even if there's noise or any small changes in V_{in}

opA forces BJT to change R s.t. V = 2V jo bhi chahiye

Series low drop out V regulator



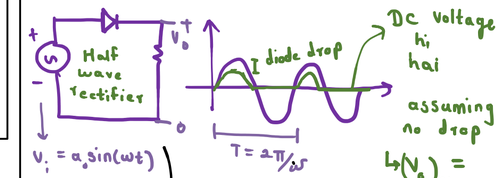
Ic if β↑↑↑

$I_{loss} = I_c(R_1+R_2) + (-V_o+V_i) \times I_c + P_{zener} + I^2R$

$V_o^2/(R_1+R_2)$, R_1, R_2 high so to 0 R_{loss}

BJT [V_i ↑ nhi dena]

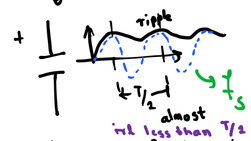
Capacitor is used along with V input for PCB.



DC voltage hi hai

assuming no drop

↳ $(V_o)_{avg}$ =



V_{ripple} = ?

f_s, T

RC >>> T

ΔV / ΔT

Full wave has half the ripple. rectifier

[0805 / 1206] → type of SMD

PCB ke upar SMD capacitor is used.

Surface mounted Device capacitor

$\Delta T = T/2$

$Q = CV$

$I = C\frac{\Delta V}{\Delta T}$

$\Delta V = \frac{I}{C}\Delta T$

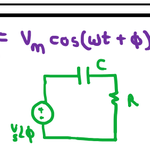
$= \frac{I}{C} \times \frac{1}{f_s} \times 2$

$\Delta V = \frac{I}{2Cf_s}$ $\quad 2f_s = f$

$\Delta V = \frac{I}{C \times f_{ripple}}$

Phasor: $V_{in} = V_m \cos(\omega t + \phi)$

Rms $\frac{V_m \angle \phi}{\sqrt{2}}$



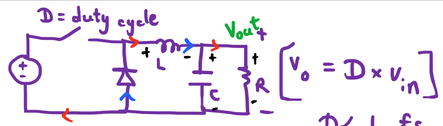
$V_{in} = \frac{V_m}{\sqrt{2}}(\cos(\omega t + \phi))$

Basically we use phasor analysis to convert DE to algebraic equat^n

$I = \int v\,dt$ gradual change so I change is gradual

$I = C\frac{dv}{dt}$ V source across C nhi cuz $I \to \infty$

D = duty cycle



$[V_o = D \times V_{in}]$

$D < 1$ f_s

assuming V_{out} to be fairly const.

ON $V_L = V_{in} - V_{out}$ } ON DT

OFF $V_L = -V_{out}$ } OFF (1−DT)

$i_{avg} = 0$

$V_{avg} = 0$

$$\text{Avg } V_L = \frac{(V_i - V_o) \times D\cancel{T} + (-V_o)(1-D)\cancel{T}}{\cancel{T}}$$

$0 = DV_i - V_o$

$[V_o = DV_i]$

Buck Converter, Boost, Buck-Boost

Study